

Sam Toughan

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About Me

Final year Computer Science student at the University of Bristol on a First Class trajectory, ranked in the top 8% of the cohort with a 75% average. Ranked first in the cohort in Image Processing and Computer Vision with 90%, and achieved 78% in Machine Learning. Currently conducting undergraduate research in machine learning, with dissertation work focused on cybernetics and human computer interaction.

Education and Qualifications

2023-2026	University of Bristol	Computer Science (BSc) Expected First-Class Honours
	A-Levels	A* in Maths, Computer Science and Physics
Work Experience	University of Bristol, in collaboration with the NHS	
Feb-26 – Current	Principal Investigator	- Building machine learning models to simulate responses to urban interventions using environmental and mobility data - Integrating green space, mobility, health, and carbon datasets for climate and wellbeing analysis - Designing a metric to quantify the combined climate and health impact of urban interventions
Aug-25 – Current	University of Bristol Graduate Teacher	- Delivered first year seminars in functional programming, imperative programming, and algorithms - Ran Computer Systems labs in C and operating systems, grading coursework and giving consistent feedback - Led Software Tools labs on Git, debugging, testing, and environment setup, producing student guides
Jun-25 – Aug-25	Vodafone AI & Automation Engineer Intern	- Automated weekly and monthly reporting, reducing preparation time from 2 hours to 30 minutes with validation checks - Built ETL pipelines from internal systems and APIs, adding anomaly flags and owner notifications - Introduced monitoring and runbooks to improve automation reliability and operational support
May-25 – Jun-25	University of Bristol Software Engineer Intern	- Co built an interactive programming game for Bristol CS Open Days using JavaScript based controls - Built the React frontend, browser based parser, and PostgreSQL backed APIs, then deployed the platform to the cloud - Used by 300+ visitors, with telemetry driven changes reducing level completion errors by 25%
Dec-24 – Mar-25	IBM Software Engineer (Contract)	- Built a cross platform weather application combining live forecasts, voice interaction, and generative visual output - Developed Flutter frontend features and backend services in Flask and MongoDB, integrating external APIs for weather - Improved system robustness through error handling, fallback flows, containerised deployment, and CI/CD automation
Projects	Acquisition Spanish Learning Web App (Next.js, React, TypeScript, Supabase, Python) - Built and deployed Acquisition (languageacquisition.net), a Spanish learning PWA with flashcards, reading, and listening - Engineered a custom Kneser-Ney NLG pipeline with target anchored beam search over a 95M sentence corpus - Designed a custom heuristic spaced repetition scheduler with five outcome paths and adaptive workload sizing Lattice Boltzmann Solver Optimisation (C, OpenMP, performance profiling, HPC systems) - Optimised a D2Q9 BGK lattice Boltzmann solver in C on the Isambard 3 supercomputer - Reworked memory layout and compute kernels to improve cache efficiency, vectorisation, and OpenMP scalability - Reduced runtime from 589s to 3s on 144 cores, guided by hardware counter analysis and roofline modelling Cornell Box Rendering Project (C++ / OOP / Rasterising / Ray Tracing) - Built a Cornell Box renderer from first principles in C++ using object oriented design - Implemented physically based materials, environment mapping, depth of field, and camera models - Applied Monte Carlo path tracing with stochastic sampling for realistic lighting and reflections	
Skills:	Languages: C, C++, Python, Go, Java, Haskell, SQL, TypeScript Frameworks & Tools: React, Next.js, Flutter, Flask, Supabase, PostgreSQL, Docker, Git, CI/CD	
Interests:	Water polo (inter-university competition), marathon running, guitar performance, language learning (German and Italian), and Vice President of the University of Bristol Spanish Society	
Achievements:	Top of cohort in Image Processing & Computer Vision (90%) Ranked in top 8% of cohort (75% 17/200), BSc Computer Science Appointed Principal Investigator on a University of Bristol research project as an undergraduate student Selected as Graduate Teacher for Object Oriented Programming, Software Tools, Computer Systems, and first-year seminars	
Languages:	English (C2), Arabic (C2), Spanish (C2), French (B1), German (A2), Italian (A2)	